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COMPUTER

CHAPTER 18

Cyber Security

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COMPUTER · CHAPTER 18

Cyber Security

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IN THIS CHAPTER

- Objectives Of Cyber Security
- Cia Triad (Foundation Of Cyber Security)
- Cia Triad Comparison
- Cyber Attack
- Types Of Cyber Attacks
- Web Attacks Summary

Cyber Security is the practice of protecting computers, mobile devices, servers, networks, software, and data from cyber attacks, unauthorized access, damage, theft, or misuse.

It helps protect:

- Hardware
- Software
- Networks
- Databases
- Personal Information
- Financial Information

Real-Life Example

When you use:

- Internet Banking
- UPI Apps
- Social Media
- E-mail

Cyber Security protects your data from hackers and cyber criminals.

Important Exam Fact

Many study materials mention **August Kerckhoffs (Auguste Kerckhoffs)** in relation to security principles. He is famous for **Kerckhoffs's Principle** in cryptography.However, there is no universally accepted "Father of Cyber Security."

OBJECTIVES OF CYBER SECURITY

Cyber Security

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├── Protect Data

├── Protect Networks

├── Protect Systems

├── Prevent Cyber Crimes

└── Ensure Safe Communication

CIA TRIAD (FOUNDATION OF CYBER SECURITY)

The CIA Triad is the fundamental model of Cyber Security.

CIA stands for:

C → Confidentiality

I → Integrity

A → Availability

A. CONFIDENTIALITY

Confidentiality means ensuring that information is accessible only to authorized users.

It prevents unauthorized access and disclosure of data.

Example

Only a bank customer should be able to view his bank account details.

Methods to Maintain Confidentiality

- Data Encryption
- Password Protection
- Two-Factor Authentication (2FA)
- Biometric Authentication
- Security Tokens

Exam Point

✔ Confidentiality = Privacy of Data

B. INTEGRITY

Integrity means ensuring that data remains accurate and unchanged.

No unauthorized person should modify the information.

Example

A student's examination result should not be altered by unauthorized users.

Methods to Maintain Integrity

- Cryptographic Checksums
- Digital Signatures
- File Permissions
- Data Backups
- Uninterruptible Power Supply (UPS)

Exam Point

✔ Integrity = Accuracy of Data

C. AVAILABILITY

Availability ensures that authorized users can access information whenever required.

Example

Online banking servers should remain available 24×7.

Methods to Maintain Availability

- Data Redundancy
- Backup Servers
- Firewalls
- Backup Power Supplies
- Disaster Recovery Systems

Exam Point

✔ Availability = Accessibility of Data

CIA TRIAD COMPARISON

Principle	Meaning	Example
Confidentiality	Prevent unauthorized access	Password
Integrity	Prevent unauthorized modification	Digital Signature
Availability	Ensure access when needed	Backup Server

CYBER ATTACK

A Cyber Attack is an attempt to gain unauthorized access to a computer system, network, or data.

The objective may be:

- Data Theft
- Financial Fraud
- Service Disruption
- Identity Theft
- Espionage

TYPES OF CYBER ATTACKS

Cyber Attacks

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├── Web-Based Attacks

└── System-Based Attacks

A. WEB-BASED ATTACKS

These attacks target websites and web applications.

1. INJECTION ATTACK

An attacker inserts malicious code into a web application to access or manipulate data.

Examples

- SQL Injection
- Code Injection
- XML Injection
- Log Injection

Exam Point

✔ SQL Injection is the most commonly asked example.

2. DNS SPOOFING

An attacker manipulates DNS records and redirects users to a fake website.

Example

Typing: www.bank.com but opening a fake website controlled by attackers.

Exam Point

✔ DNS Spoofing redirects users to wrong IP addresses.

3. SESSION HIJACKING

An attacker steals a user's session ID or cookies and gains unauthorized access.

Example

Stealing login session of a social media account.

Exam Point

✔ Session Hijacking often involves cookie theft.

4. PHISHING

A fraudulent attempt to obtain sensitive information by pretending to be a trusted organization.

Example

Fake bank email asking you to verify account details.

Exam Point

✔ Phishing is one of the most common cyber attacks.

5. BRUTE FORCE ATTACK

A trial-and-error attack used to guess passwords.

Example

Trying thousands of password combinations automatically.

Exam Point

✔ Brute Force = Password Guessing Attack

6. DENIAL OF SERVICE (DoS)

An attack that makes a website or server unavailable to legitimate users.

It floods the target system with excessive requests.

Example

Thousands of fake requests crash a website.

Exam Point

✔ DoS uses a single system to attack.

7. DICTIONARY ATTACK

An attacker uses a list of commonly used passwords.

Examples

- 123456
- password
- admin123

Difference from Brute Force

Dictionary Attack uses a predefined list.

Brute Force tries all possible combinations.

8. URL INTERPRETATION ATTACK

An attacker modifies a URL to access unauthorized pages.

Example :Changing:/user/profile to /admin/profile to gain unauthorized access.

9. FILE INCLUSION ATTACK

Allows attackers to access sensitive files or execute malicious files on a web server.

Exam Point

✔ Exploits file inclusion functionality.

10. MAN-IN-THE-MIDDLE (MITM) ATTACK

An attacker secretly intercepts communication between two parties.

Example

Public Wi-Fi interception.

Exam Point

✔ MITM acts as a hidden bridge between sender and receiver.

WEB ATTACKS SUMMARY

Attack	Purpose
SQL Injection	Access Database
DNS Spoofing	Redirect Traffic
Session Hijacking	Steal Session
Phishing	Steal Credentials
Brute Force	Guess Password

Attack	Purpose
DoS	Make Service Unavailable
Dictionary Attack	Use Common Passwords
URL Interpretation	Unauthorized Access
File Inclusion	Access Sensitive Files
MITM	Intercept Communication

B. SYSTEM-BASED ATTACKS

These attacks target computers and networks.

1. VIRUS

A Virus is malicious software that attaches itself to files and spreads when executed.

Characteristics

- Requires user action
- Self-replicates
- Damages files

Exam Point

✔ Virus needs a host file.

2. WORM

A Worm is malware that spreads automatically across networks.

Characteristics

- Self-replicating
- Does not require a host file

Exam Point

✔ Worm spreads without user intervention.

3. TROJAN HORSE

A malicious program disguised as legitimate software.

Example

Fake antivirus software.

Exam Point

✔ Trojan appears genuine but performs malicious actions.

4. BACKDOOR

A hidden method that bypasses normal authentication mechanisms.

Example

Secret access built into software.

5. BOT

A Bot is an automated software program.

Malicious Examples

- Botnets
- Spam Bots

THE 7 LAYERS OF CYBER SECURITY

7. Human Layer

6. Perimeter Security

5. Network Security

4. Endpoint Security

3. Application Security

2. Data Security

1. Mission Critical Assets

1. Mission Critical Assets

Most valuable organizational assets.

Examples:

- Customer Database
- Financial Records

2. Data Security

Protects data during storage and transmission.

3. Application Security

Protects software applications

4. Endpoint Security

Protects devices such as:

- Computers
- Smartphones
- Tablets

5. Network Security

Protects organizational networks from attacks.

6. Perimeter Security

Protects organizational boundaries.

Examples:

- Firewalls
- Physical Security

7. Human Layer

Protects against human mistakes and social engineering attacks.

Exam Point

- ✔ Humans are considered the weakest link in cyber security.

TYPES OF ATTACKERS

1. PASSIVE ATTACKER

Monitors and collects information without changing system resources.

Example :Eavesdropping on communication.

2. ACTIVE ATTACKER

Modifies data or disrupts system operations.

Example

Deleting files or altering data.

3. INSIDER ATTACKER

Attack initiated from within the organization.

Example

Disgruntled employee stealing data.

4. OUTSIDER ATTACKER

Attack initiated from outside the organization.

Example

External hacker attacking company servers.

IMPORTANT CYBER SECURITY TERMS

FIREWALL

A security system that monitors and controls incoming and outgoing network traffic.

Exam Point

- ✔ Firewall acts as a barrier between trusted and untrusted networks.

ANTIVIRUS

Software designed to detect and remove malware.

Examples:

- Quick Heal
- Norton AntiVirus
- McAfee

ENCRYPTION

Process of converting readable data into unreadable form.

Example

HELLO → Encrypted Text

DECRYPTION

Converting encrypted data back into readable form.

QUICK REVISION NOTES

CIA Triad

- C → Confidentiality
- I → Integrity
- A → Availability

Malware Types

- Virus
- Worm
- Trojan
- Bot

Common Attacks

- Phishing
- SQL Injection
- DoS
- MITM
- DNS Spoofing

Attackers

- Passive
- Active
- Insider
- Outsider

MOST IMPORTANT FACTS FOR EXAMS

1. CIA Triad is the foundation of Cyber Security.

2. Confidentiality protects privacy.

3. Integrity protects accuracy.

4. Availability ensures accessibility.

5. Phishing steals sensitive information.

6. SQL Injection targets databases.

7. DNS Spoofing redirects users to fake websites.

8. Session Hijacking steals user sessions.

9. DoS makes services unavailable.

10. Virus requires a host file.

11. Worm spreads independently.

12. Trojan disguises itself as legitimate software.

13. Humans are the weakest link in cyber security.

14. Firewall protects networks from unauthorized access.

15. Encryption converts plaintext into ciphertext.